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$ cat 12-payments-subscription.md

MAJIBILL PLATFORM · MONEY FLOWS

Payments & Subscription Billing

Two money flows kept strictly apart: customers paying utilities, and utilities paying the platform — plans, usage metering, invoicing, dunning, entitlements and partner settlement.

```
usage = count(active billable accounts) per tenant per month
// customer money never touches the platform
// suspended tenants still receive customer payments
```

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// Contents

1. Two money flows . . . 3
2. Flow A — customer payments . . . 3
3. Flow B — subscription model . . . 5
4. Usage metering and invoicing . . . 6
5. Dunning, suspension and reinstatement . . . 7
6. Partner margin and settlement . . . 7
7. Data model (platform database) . . . 8
8. Platform console views . . . 8
9. Edge cases and decisions . . . 9

// 1. Two money flows

MajiBill involves two completely different flows of money. Confusing them is the most common design mistake in platforms like this. **Flow A** is a utility's customers paying water bills; the platform records it but does not hold it. **Flow B** is the utility (tenant) paying 404HubSpot for the platform, with a share going to ZUCAR as partner.

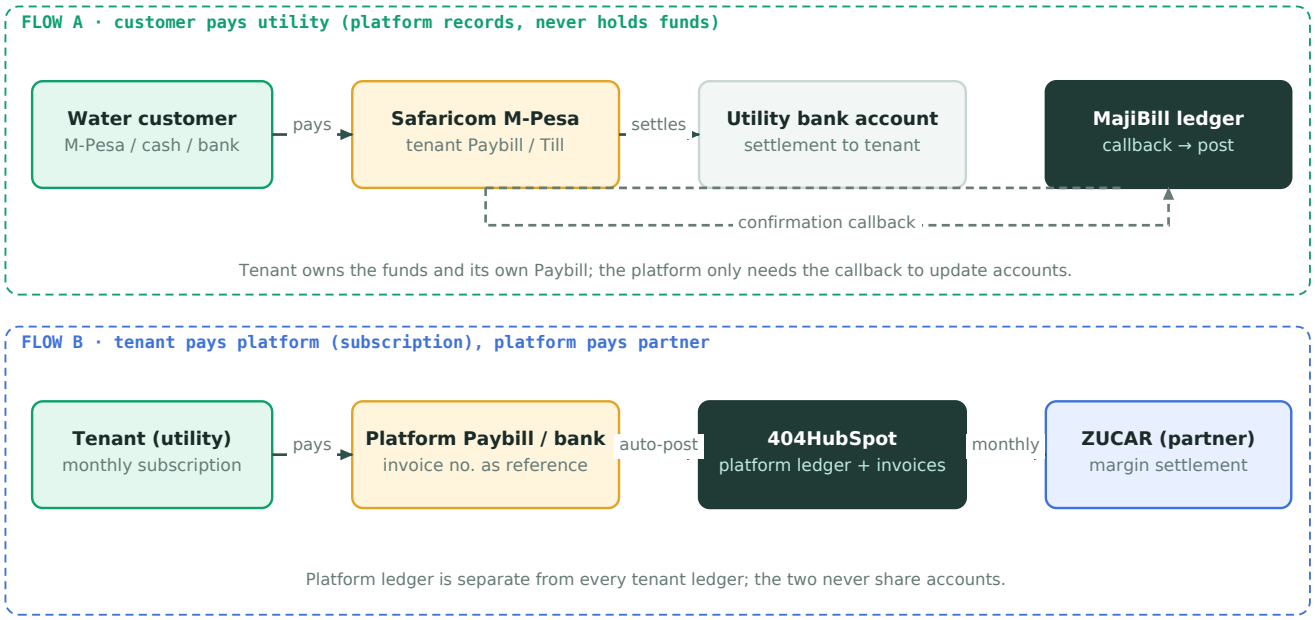


Fig 1.1 Flow A (top) is tenant money. Flow B (bottom) is platform revenue.

> DECISION **Platform never holds customer funds**
> Each tenant uses its own Paybill/Till so settlement goes straight to the utility. This keeps the platform out of custody of customer money. Confirm with Safaricom and counsel before offering any shared or aggregated collection or wallet feature (MB-07).

// 2. Flow A — customer payments

Channel	How it works	Posting
M-Pesa C2B Paybill	Customer pays Paybill with account number; validation and confirmation callbacks to the Go ingress (MB-01)	Automatic, within seconds
M-Pesa STK push	Portal or clerk triggers a prompt on the customer's phone; result via callback	Automatic; linked to the request
Cash / cheque at office	Cashier session records payment and issues receipt; end-of-day close-out	Immediate, supervisor close-out
Bank deposit / transfer	Bank statement CSV import (later bank APIs); matched by reference	Batch, with suggestions for unmatched
Prepayment / credit	Overpayments and advance payments held as credit on the account	Auto-applied to next bill

2.1 M-Pesa integration checklist (per tenant)

1. Tenant obtains Paybill/Till and Daraja credentials; platform stores them in the secrets manager, never in the database or logs.
2. Register validation and confirmation URLs per shortcode; each URL path carries an unguessable token.
3. Test in sandbox; run the production go-live process with Safaricom (timelines are outside our control; start in week 1).
4. Run a live test payment end to end and confirm receipt SMS and ledger entries.

5. Enable daily reconciliation against statements/transaction-status queries; set variance alert thresholds.
6. Document the runbook: callback outage, replay from inbox, reversals, fee handling.

- Reversals and disputes arrive as events and create **reversal journals**, never edits (MB-03).
- Transaction fees borne by customer or utility are tenant configuration; the ledger records the gross payment and any fee entry separately.
- Payment exceptions queue (suspense) has an owner, ageing and SLA; unresolved items older than a threshold alert the finance manager.

// 3. Flow B — subscription model

3.1 What is billed

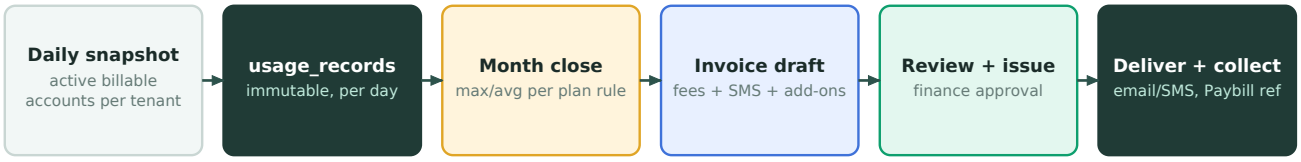
Component	Basis	Notes
Platform fee	Per **active billable connection** per month, tiered	Active = account billed in the month; closed/disconnected-no-bill excluded by plan rule
Minimum monthly fee	Fixed	Protects against tiny tenants costing more than they pay
Onboarding / setup	One-time per tenant	Data migration, training, configuration
SMS	Pass-through at cost + handling, or bundle with cap	Per-tenant SMS wallet with low-balance alerts
Add-on modules	Monthly or one-time	Reader app seats, STK push, analytics, API access, SSO
Dedicated cell premium	Monthly	Silo infrastructure, dedicated key management, in-country hosting
Support tier	Monthly	Standard / Extended / 24x7 (MB-10)

3.2 Plans and entitlements (structure; prices set in the commercial model)

Capability	Starter	Standard	Enterprise
Connections included / cap	up to 5,000	up to 100,000	unlimited, dedicated cell
Customers, meters, billing, ledger, reports	Y	Y	Y
M-Pesa auto-posting	Y	Y	Y
Offline reader app	Y (limited seats)	Y	Y
SMS notifications	pay-as-you-go	bundle + pay-as-you-go	negotiated
Customer portal + STK push	—	Y	Y
API and webhooks	—	Y	Y
SSO (OIDC/SAML)	—	—	Y
Advanced analytics (NRW, anomalies)	—	add-on	Y
Audit export and integrity reports	basic	Y	Y + auditor access
Support	Standard	Standard / Extended	Extended / 24x7
In-country / dedicated hosting	—	—	optional

Entitlements are stored as feature flags and numeric limits on the subscription and read by the same policy function as RBAC (MB-11). Limits are **soft first** (warn at 80% and 100%) then hard only where it protects the platform (e.g. SMS wallet at zero, connection cap on Starter).

// 4. Usage metering and invoicing



Usage is measured from the tenant's own ledger and bill data, so the invoice can be audited the same way as a utility bill.

Fig 4.1 Metering to invoice pipeline.

- **Billing basis rule per plan:** peak, average, or month-end count of active billable connections; stated in the contract and shown on the invoice.
- **Proration:** upgrades take effect immediately with prorated charges; downgrades at next period start. Pilot and trial periods are explicit subscription phases with their own price.
- **Credit notes** for errors or service credits; invoices are never edited after issue.
- **Currency:** KES by default; invoices show tax lines according to the platform company's tax position. Confirm VAT and electronic tax-invoice obligations with an accountant once registered.

// 5. Dunning, suspension and reinstatement

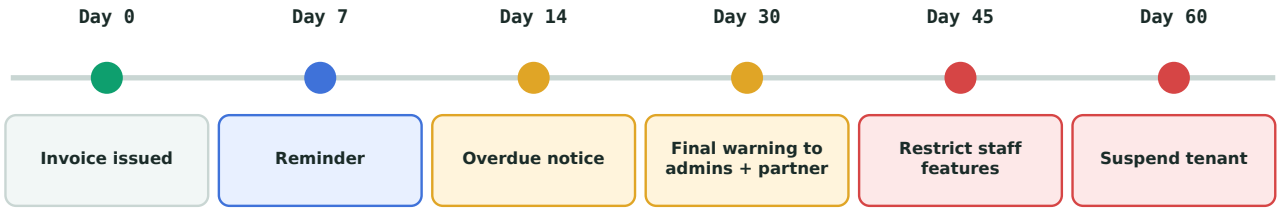


Fig 5.1 Default dunning schedule (configurable per plan and contract).

Stage	Platform behaviour
Reminder / overdue	Email and in-app banner to tenant admins; partner copied
Final warning	Personal call or message from account manager; payment plan option
Restricted	Read-only for non-essential modules; **billing runs already scheduled complete**; payments still post
Suspended	Staff sign-in blocked except export and billing-contact access; customer payments still ingest and post; SMS paused except receipts if wallet allows
Reinstatement	On payment (auto-matched by invoice reference) the tenant returns to Active within minutes; audit event records the change

5.1 Subscription payment methods

- Platform Paybill with the **invoice number as account reference**, auto-reconciled to the invoice by the same inbox/ledger machinery used for customer payments (on the platform ledger).
- Bank transfer with statement import; card via a hosted, PCI-compliant checkout (no card data stored).
- Annual prepay with discount; purchase orders for institutional tenants; partner-collected invoicing where ZUCAR bills the end customer and pays the platform (agreement-specific).

// 6. Partner margin and settlement

Step	Detail
Attribution	Each tenant has a `partner_id` and a margin rule (percentage, tiering, duration) in the contract record
Calculation	Monthly, on **collected** subscription revenue (or invoiced, per contract), excluding pass-through costs and taxes
Statement	Partner statement lists tenants, usage, invoiced, collected, margin, adjustments; visible in the partner portal
Settlement	Payout via bank transfer or M-Pesa B2C after approval (maker–checker); payout journal in the platform ledger
Clawbacks	Refunds and credit notes reverse margin proportionally in the next statement
Audit	Statements are immutable, hashed, and covered by the platform audit chain (MB-06)

Worked example, **hypothetical numbers**: a tenant with 30,000 billable connections at KES 20 per connection is KES 600,000 per month. With a 25% partner margin the partner statement shows KES 150,000 and the platform retains KES 450,000, before SMS and other pass-throughs that are billed at cost.

// 7. Data model (platform database)

```sql (platform database)
CREATE TABLE subscriptions (
  id uuid PRIMARY KEY, tenant_id uuid NOT NULL UNIQUE, plan_id uuid NOT NULL,
  status text CHECK (status IN ('trial','pilot','active','past_due','restricted','suspended','cancelled')),
  billing_basis text NOT NULL,            -- 'peak' | 'average' | 'month_end'
  period_start date NOT NULL, current_period_end date NOT NULL,
  entitlements jsonb NOT NULL,            -- flags + numeric limits
  partner_margin jsonb                    -- rule snapshot
);
CREATE TABLE usage_records (
  tenant_id uuid NOT NULL, day date NOT NULL, metric text NOT NULL, value bigint NOT NULL,
  PRIMARY KEY (tenant_id, day, metric)                  -- immutable once month is closed
);
CREATE TABLE platform_invoices (
  id uuid PRIMARY KEY, tenant_id uuid NOT NULL, invoice_no text UNIQUE NOT NULL,
  period daterange NOT NULL, total_minor bigint NOT NULL, currency char(3) DEFAULT 'KES',
  status text CHECK (status IN ('draft','issued','paid','overdue','void','written_off')),
  issued_at timestamptz, due_at date
);
```

Table	Purpose
`plans`, `plan_prices`	Catalogue, tiers, effective-dated prices
`subscriptions`	Tenant's current plan, status, entitlements, margin rule
`usage_records`	Daily metrics (billable connections, SMS segments, storage, seats)
`platform_invoices`, `invoice_lines`	Issued documents; never edited
`platform_payments`, `platform_allocations`	Payments received for invoices (own inbox + ledger)
`partner_statements`, `payouts`	Monthly partner settlement
`credit_notes`	Corrections and service credits
`platform_ledger_*`	Separate chart of accounts for the platform company

// 8. Platform console views

View	Content
Revenue	MRR/ARR, new, expansion, contraction, churn, ARPA, collection rate, ageing of subscription receivables
Tenants	Plan, usage vs entitlement, invoice status, health score, support tier
Partners	Tenants sourced, margin accrued, statements, payouts
Cost and margin	Cell cost per tenant (tags), SMS cost vs recharge, gross margin by tenant (MB-05)
Exceptions	Failed usage snapshots, invoices without payment reference, entitlement breaches

// 9. Edge cases and decisions

Case	Decision
Tenant disputes usage	Provide the per-day usage report and the underlying account list; credit note if wrong
Tenant pays the wrong reference	Suspense on the platform ledger; finance resolves; no auto-guess above a risk threshold
Plan cap exceeded mid-month	Soft limit: warn and bill overage per plan; hard limit only for Starter’s cap by contract
Partner leaves	Tenant remains with the platform; margin ends or transitions per agreement; no service interruption
Tenant migrates to sovereign licence	Subscription converts to licence + support; usage metering ends; licence key issued
Currency or price change	Effective-dated price versions; existing contracts keep their price until renewal
Refund of a subscription payment	Credit note first; cash refund by approval; reversal journal on platform ledger